PABITRA KUMAR ROY

MS in Transportation Engineering
Portland State University, Oregon, USA
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PROFESSIONAL PROFILE

Transportation Engineering with over 8 years of experience bridging major infrastructure and advanced traffic safety research and planning to attend FE certification (June, 2026). Proven ability to oversee major infrastructure projects and perform advanced data analysis in traffic safety and modeling. Seeking to leverage expertise in traffic impact studies, safety analysis, and project management to enhance traffic engineering outcomes.

EXPERIENCE

Graduate Research Assistant (Transportation Engineering) 01/2024 – 12/2025
Portland State University *Portland, OR, USA*

- Research focused on predictive crash modeling and driver behavior to inform traffic safety countermeasures and intersection design improvements, utilizing machine learning and econometric techniques.
- Apply spatial data science and advanced Machine Learning modeling to analyze transportation safety data.
- Utilize a robust technical toolkit including Python, R, SPSS, ArcGIS, MicroStation, and AutoCAD.
- Analysis complex traffic safety research into instructional materials, mentoring students in transportation engineering principles.

Office Engineer / Site Supervision Engineer (Tunnel Project) 09/2018 – 12/2023
SMEC International Pty Ltd. *Bangladesh / China*

- Served as Site Supervision Engineer for the landmark Bangabandhu Sheikh Mujibur Rahman Tunnel (a multi-lane road tunnel under the Karnaphuli River), representing client interests and ensuring contractor compliance with design specifications and quality standards.
- Provided full-time international supervision at the Zhenjiang prefabrication factory in China, overseeing quality assurance for the production of all 20,110 precast tunnel segments (12.12m diameter), including rebar fabrication, mechanical testing (leakage, bending, pull-out), and managing safe transportation logistics to Bangladesh.
- Supervised the construction of twin 2.5km slurry shield tunnels using Tunnel Boring Machines (TBM), monitoring all activities from assembly and commissioning to excavation, slurry quality control, segment installation, and cross-passage construction.
- Directed civil works for the 727-meter viaduct, monitoring bored pile construction, pier columns, bent caps, and the casting, pre-stressing, and installation of precast box girders, ensuring adherence to strict engineering standards.
- Managed geotechnical and pavement activities for 5km of 4-lane roadway, including subgrade improvement via PVD installation, soil stabilization, preloading, and the construction of flexible/rigid pavements, culverts, and underpasses.
- Oversaw structural works for all ancillary facilities, including the Weigh Station, Toll Plaza (9 buildings), and Service Area, monitoring foundations, superstructure erection, and architectural concrete work.
- Enforced rigorous Quality Control/Quality Assurance (QC/QA) protocols across all project phases, witnessing laboratory and field tests, inspecting materials (concrete, rebar), and approving all work prior to acceptance.
- Executed comprehensive project coordination and reporting, including preparing daily/monthly progress reports, technical documentation, subcontractor billing, and minutes of client meetings for senior management.
- Monitored and supervised construction of twin 2.5km x 12.12m diameter slurry shield tunnels using a tunnel boring machine (TBM), including assembly, commissioning, and excavation. Oversaw TBM parameters, slurry performance, grouting, segment installation, leakage and crack monitoring, cutterhead interventions, and shield deformation checks.

  *(For detailed project photos and further evidence of project scope and **Pabitra's** involvement, click here)*

Assistant Engineer (Quality Assurance & Control) 06/2017 – 09/2018
Next Spaces Limited *Bangladesh*

- Directed quality control for geotechnical construction, executing and analyzing field tests (cross-hole, plate load, pressuremeter, permeability) to validate soil parameters and foundation performance.
- Managed the full cycle of in-situ sand-cement piling, from overseeing construction to extracting, preparing, and laboratory testing core samples to verify pile integrity and design compliance.
- Spearheaded project quality and safety protocols by preparing QC reports, work plans, and schedules, while conducting weekly safety and planning meetings to maintain strict HSE standards.

- Coordinated daily site operations and resource planning, including manpower mobilization, to achieve project milestones, supported by regular progress reporting to the Project Manager.
- Supported project timeline adherence through strategic manpower planning, progress tracking, and direct daily reporting to senior management.

Assistant Engineer (Construction) 12/2015 – 06/2017
MAK Estate Development Ltd. *Bangladesh*

- Managed technical document control, reviewing all plans, drawings, and quantity calculations for accuracy before site implementation and procurement.
- Directed on-site coordination, logistics, and material supply chains, ensuring strict compliance with project specifications, budgets, and timelines.
- Conducted daily supervision, quality inspections, and safety audits to maintain design integrity and mitigate on-site risks.
- Oversaw project administration, preparing record drawings, technical reports, daily/monthly progress summaries, and subcontractor billing documentation.
- Liaised with the purchasing department to define material requirements and maintained precise inventory records of materials received, consumed, and in stock.

EDUCATION

- **MS in Transportation Engineering**, Portland State University, USA, *2025* *3.81 (4.00)*
- **BSc in Civil Engineering**, RUET, Bangladesh, *2016* *3.39 (4.00)*

SKILLS

- Programming Languages**:** Python, R, Gauss
- Software: Microsoft Office, ArcGIS, AutoCAD, MicroStation, TransCAD, SPSS, Canva, MIDAS GTSNX
- Report writing, project specifications, and cost estimates.
- Field experience in supervising various transportation projects.

LEADERSHIP/MEMBERSHIPS

Institute of Transportation Engineers (ITE-STEP), Portland State University 02/2025 - Present

- *Vice-President*
- *Managed a $20,000 club budget and prepared yearly budget plans and impact reports.*
- *Worked with the ITE leadership team to plan and run club events.*
- *Won Oregon ITE Traffic Bowl Competition (2nd Place) -2024*

Bangladesh Student Association (BSA), Portland State University 03/2024 - Present

- *Treasurer*
- *Managed a $3,000 club budget, including annual planning and preparation of budgets.*
- *Collaborated with BSA officers to organize and coordinate club events.*
- Associate Editor, Hatfield Graduate Journal of Public Affairs, PSU 10/2025 - Present
- ITE Annual Meeting and Exhibition of 2025 *(Student Reviewer)* 11/2024 – 02/2025
- Geospatial Information Society (GIS), PSU *(Student Member)* 09/2024 - Present
- Association for Commuter Transportation (ACT) *(Student Member)* 03/2024 - Present
- American Society for Civil Engineers (ASCE) *(Student Member)* 03/2024 - Present
- Institution of Engineers, Bangladesh (IEB) *(Associate Member)* 01/2020 - Present
- ASCE Transportation & Development Institute (T&DI) *(Student Member)* 09/2024 - Present

COURSES COMPLETED

- Transportation Optimization
- Public Transportation System
- Transportation Seminar
- Transportation Data Science
- Design and operation of Bike & Pedestrian Infrastructure
- Autonomous Vehicle Safety
- Transportation Safety Analysis
- Transportation Economics
- Smart Cities
- Modern Statistics for Engineers
- Econometric modeling
- Traffic Flow Theory

HONORS & AWARDS

- 2026 Clifford Spiegelman Student Paper Competition, ASA-TSIG (3rd Place) 2026
- TRB-2026 Conference Travel Grants 2025 & 2026
- Oregon ITE Traffic Bowl Competition (2nd Place as team) (2 times) 2025 & 2024
- PacTrans-2024 Conference Travel Grants 2024
- Student Leadership Fellows Scholarship 2025 & 2024